

Amazon.com, Inc.

(AMZN) · Monte Carlo DCF Valuation Report

Central intrinsic value per share

\$96.52

WACC 11.27% | Market \$246.03 (as of 2026-06-05)

Generated by the Monte Carlo Valuation Engine. Engine applied as-is; no DCF / Monte Carlo / shock / convergence logic was modified. Figures are a statement about the input assumptions, not investment advice.

1. Assumptions

Input	Value	Rationale
Starting revenue	\$716,900M	FY2025 actual net sales (+12% YoY)
Net debt	-\$57,400M	Net cash: long-term debt \$65.7B minus cash \$86.8B and marketable securities \$36.2B. Operating-lease liabilities excluded (treated as operating, not financing).
Shares outstanding	10,730M	Approx. diluted share count, FY2025
Forecast horizon	7 years	Explicit-forecast window before terminal value
Revenue growth	11% -> 6%	Fades from low-double-digit (AWS + ads led) toward mature large-cap single digits
Operating margin	12% -> 15%	Gradual expansion as higher-margin AWS and advertising grow as a share of mix
Capex % revenue	16% -> 9%	Elevated AI/datacenter buildout now (FY2025 capex >\$100B; management guided ~\$200B for 2026), normalising over the horizon. This is the fixture's most aggressive and most uncertain assumption.
D&A % revenue	5.8% -> 7.0%	Rising as the AI-era asset base grows and server useful lives shorten
NWC % revenue	0.0% flat	Conservatively neutral; Amazon historically runs negative working capital (supplier-financed), so this understates a genuine cash advantage
Tax rate	15%	Near recent effective rate
Terminal growth	2.5%	Long-run nominal GDP-like
Risk-free rate	4.3%	Long Treasury
Equity risk premium	5.5%	Standard equity premium
Beta	1.30	Above-market systematic risk; published readings ranged ~1.3-1.5, the lower end chosen as a central view
Cost of debt	5.0%	High-grade issuer at current rates
Debt / total capital	2.5%	Minimal financial leverage relative to ~\$2.6T market cap

These are analyst assumptions derived from FY2025 actuals and a deliberately conservative central view - not a forecast and not investment advice. The capex trajectory in particular smooths over a near-term spike (management's ~\$200B 2026 guide implies capex well above the 16% starting point); the Monte Carlo layer widens this band, but the central fixture stays conservative.

2. Deterministic Central Case

Forecast window: base year FY2025 (reported actuals); explicit forecast FY2026–FY2032 (7 years); terminal value from FY2033 onward.

WACC	11.270%
Terminal spread (WACC – g)	8.77%
Central DCF per share	\$96.52

Year-by-year assumptions

Year	Rev growth	Op margin	Capex %	D&A %	ΔNWC %
FY2026	11%	12.0%	16.0%	5.8%	0.0%
FY2027	10%	12.5%	15.0%	6.0%	0.0%
FY2028	9%	13.0%	14.0%	6.2%	0.0%
FY2029	8%	13.5%	12.0%	6.4%	0.0%
FY2030	7%	14.0%	11.0%	6.6%	0.0%
FY2031	6%	14.5%	10.0%	6.8%	0.0%
FY2032	6%	15.0%	9.0%	7.0%	0.0%

Free-cash-flow (FCFF) trajectory — USD millions

Year	Revenue	EBIT	NOPAT	+D&A	–Capex	FCFF
FY2025	\$716,900	—	—	—	—	—
FY2026	\$795,759	\$95,491	\$81,167	\$46,154	(\$127,321)	\$0
FY2027	\$875,335	\$109,417	\$93,004	\$52,520	(\$131,300)	\$14,224
FY2028	\$954,115	\$124,035	\$105,430	\$59,155	(\$133,576)	\$31,009
FY2029	\$1,030,444	\$139,110	\$118,243	\$65,948	(\$123,653)	\$60,539
FY2030	\$1,102,575	\$154,361	\$131,206	\$72,770	(\$121,283)	\$82,693
FY2031	\$1,168,730	\$169,466	\$144,046	\$79,474	(\$116,873)	\$106,647
FY2032	\$1,238,854	\$185,828	\$157,954	\$86,720	(\$111,497)	\$133,177

3. Continuous Monte Carlo (z*)

Convergence resolved at **z* = 2,000** paths (decision margin 83.2%; z_pct 1,000, z_elbow 2,000).

Mean	\$98.37
Median	\$95.82
Std deviation	\$29.98
Min / Max	\$8.36 / \$223.71
Fraction negative	0.0%

Percentile	P5	P10	P25	P50	P75	P90	P95
Value	\$53.77	\$62.78	\$76.79	\$95.82	\$117.60	\$138.98	\$151.94

Benchmark vs. the 10,000-path folk number: z* uses **0.2×** the compute yet the mean differs by only **0.9%** (\$98.37 vs \$97.48).

4. Shocked Monte Carlo (z**)

Re-converged under the discrete shock overlay at **z** = 2,000** (decision margin 36.2%).

Mean	\$90.86
Median	\$88.64
Std deviation	\$32.48
Min / Max	\$-1.44 / \$234.12
Fraction negative	0.1%

Percentile	P5	P10	P25	P50	P75	P90	P95
Value	\$40.20	\$52.04	\$68.39	\$88.64	\$110.26	\$132.14	\$146.08

Shocks shift the whole distribution down (shocked mean ~\$90.9 vs continuous ~\$98.4) and fatten the left tail (P5 falls from ~\$53.8 to ~\$40.2, min from \$8.36 to slightly negative). A small fraction of shocked paths (0.05%) cross into negative intrinsic value - Amazon's thinner GAAP margins and heavy ongoing investment leave less cushion than a net-cash, high-margin software peer.

5. Shock Channel Behaviour

Channel	Revenue	Margin	Funding	Strategic	Cash
Fires — all paths	448	417	436	415	453
Fires — worst 5%	97	127	44	52	85

Shock-free paths: **67.4%**. In the worst-5% tail, margin (127) and revenue (97) dominate, with cash (85) also elevated, while funding and strategic sit near their base rate. The downside is an operating story - margin

compression and demand loss during a heavy-investment phase - rather than a financing story, consistent with the V1 fragility thesis for a company carrying little financial debt but large fixed-investment commitments.

6. Market Comparison

Amazon.com, Inc. traded at **\$246.03** on 2026-06-05. Against the continuous distribution the market price sits above the maximum simulated value (\$223.71). Against the shocked distribution it sits above the maximum simulated value (\$234.12).

The market price (~\$246) sits ABOVE the entire simulated distribution under both engines - the continuous run never produces a value above ~\$224, the shocked run never above ~\$234. On this conservative fixture the engine literally cannot manufacture today's price: the central intrinsic value is ~\$96.52, roughly 61% below market. The gap is the market's implied bet that capex converts into far higher future cash flow (AI/AWS operating leverage) and/or that margins expand faster than the fixture assumes. The model cleanly isolates 'what these conservative fundamentals support' from 'what the market is paying for the AI build-out story.'

7. Seed Robustness

Seed	z* (cont)	cont margin	z** (shock)	shock margin
42	2000	83.2%	2000	36.2%
99	3000	101.5%	1500	3.0%
123	1500	25.0%	1500	61.5%

Across seeds, z* (continuous) ranges 1,500-3,000 and z** (shocked) 1,500-2,000 - all comfortably within the 10,000 grid. As with MSFT, the point estimate of the required sample size is seed-sensitive (an elbow/percentile rule on finite batches), but the qualitative conclusion is stable: Amazon is a tractable convergence problem resolved well below the folk number on every seed. One shocked seed (99) flagged a very tight margin (3.0%) with a large batch recommendation, a reminder that the shocked stage is the harder convergence problem.

8. Conclusion

On a conservative FY2025-base fixture, Amazon's Monte Carlo intrinsic value is a distribution centred near \$96/share - continuous mean ~\$98.4, shocked mean ~\$90.9 - with almost no negative paths (0.0% continuous, 0.05% shocked). The continuous stage resolves cheaply (z* = 2,000, 83% decision margin, reproducing the 10,000-path mean within 0.9% at one-fifth the compute); the shocked stage is harder (z** = 2,000, 36% margin) with an operating-channel-driven tail. The headline result is valuation: at ~\$246 the market trades above the engine's entire simulated universe under both engines. The model, on these inputs, simply cannot reach today's price - the market is paying for an AI-capex-to-cash-flow conversion and margin trajectory more optimistic than this fixture encodes. As a real-world test, Amazon also illustrates the documented fragility of the capex-%-of-revenue assumption: a company spending >\$100B/year on long-lived assets is precisely the case where that scaling assumption is least comfortable, and where the analyst's judgment on the capex fade does most of the work in the valuation.

Engine applied as-is; no logic modified. Seed 42. Market price as of 2026-06-05. This document reports model output under stated assumptions and is not investment advice.